

Practice Quiz: Planning (Computational Neuroscience)

Name: __ Date: __

Part A: Multiple Choice

1. The default mode network (DMN) is most active when: A) Performing a demanding cognitive task B) The mind is wandering, imagining future scenarios, or recalling memories C) Exercising vigorously D) Sleeping deeply
2. The hippocampus supports planning by: A) Only storing exact copies of past events B) Recombining memory elements to construct novel future scenarios C) Directly controlling muscle movements D) Processing visual information
3. Goal-directed behavior is primarily implemented by: A) The putamen B) The prefrontal cortex → caudate nucleus circuit C) The cerebellum D) The brainstem
4. Habitual behavior is primarily implemented by: A) The prefrontal cortex B) The sensorimotor cortex → putamen circuit C) The hippocampus D) The amygdala
5. Time cells in the hippocampus: A) Keep track of the current time of day B) Fire at specific moments in a temporal sequence, creating a neural timeline C) Only fire during sleep D) Measure the speed of neural processing
6. Patients with hippocampal damage: A) Can imagine the future normally but can't remember the past B) Have impaired ability to both remember past events and imagine future scenarios C) Only lose short-term memory D) Develop enhanced planning abilities
7. The shift from goal-directed to habitual behavior occurs when: A) The agent becomes lazy B) Prediction errors decrease with practice, making strong policy priors sufficient C) The prefrontal cortex grows new neurons D) Dopamine is eliminated

Part B: Short Answer

1. Explain why the DMN is sometimes called the brain's "simulator." What does it simulate, and how does this relate to planning as inference in Active Inference?
2. A person with depression engages in rumination — repeatedly imagining negative future scenarios. Using the DMN and constructive memory framework, explain how rumination might be understood as maladaptive offline simulation. What might be different about the generative model?
3. Describe how the same hippocampal neuron populations could support both memory retrieval and future imagination. Why does this shared architecture make evolutionary sense?